

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 23 — Chemical Energetics (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Comprehensive Multiple Choice Questions · Quick-Check Answer Grid · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Topic 23 MCQs (100 Core Questions)**High-Frequency Core Repeats (Q101 – Q110)**

- Comprehensive coverage across lattice energy, Born-Haber cycles, hydration and solution thermodynamics
- The 10 most frequently examined Cambridge Paper 1 questions on A Level Chemical Energetics.

SUBTOPIC 23.1 — LATTICE ENERGY & BORN-HABER CYCLES (Q1 – Q25)

1 Definition of Lattice Energy — 9701/11/M/J/23/Q1

EASY · Syllabus 23.1

Which statement correctly defines the standard lattice energy of an ionic compound?

- A: The energy required to break one mole of an ionic compound into its constituent gaseous atoms.
- B: The enthalpy change when one mole of an ionic compound is formed from its constituent gaseous ions under standard conditions.
- C: The enthalpy change when one mole of an ionic compound dissolves in water to form an infinitely dilute solution.
- D: The energy required to remove one mole of electrons from one mole of gaseous cations.

2 Lattice Energy Comparison — 9701/12/M/J/23/Q1

EASY · Syllabus 23.1

Which ionic compound has the most exothermic lattice energy?

- A: NaCl
- B: MgCl₂
- C: BaO
- D: MgO

3 Second Electron Affinity of Oxygen — 9701/13/M/J/23/Q1

HARD · Syllabus 23.1

Why is the second electron affinity of oxygen, $O^-(g) + e^- \rightarrow O^{2-}(g)$, endothermic (+798 kJ mol⁻¹)?

- A: Energy must be supplied to overcome electrostatic repulsion between the incoming electron and the negative O⁻ ion.
- B: Oxygen achieves a stable noble gas configuration.
- C: The incoming electron enters a higher principal quantum shell.
- D: The nuclear charge of oxygen decreases.

4 Theoretical vs Experimental Lattice Energies — 9701/11/O/N/23/Q1

HARD · Syllabus 23.1

The experimental lattice energy of silver iodide (–890 kJ mol⁻¹) is significantly more exothermic than its theoretical value (–699 kJ mol⁻¹). What causes this discrepancy?

- A: Silver iodide forms a giant covalent macromolecular network like diamond.
- B: Iodide ions undergo spontaneous oxidation to iodine molecules.
- C: Polarisation of the large iodide anion by the silver cation introduces significant covalent bonding character.
- D: The theoretical calculation neglects the hydration enthalpy of silver ions.

5 Born-Haber Cycle Unknown Term — 9701/12/O/N/23/Q1

HARD · Syllabus 23.1

In a Born-Haber cycle for calcium chloride, CaCl₂, which term must be multiplied by two?

- A: The first electron affinity of chlorine.
- B: The standard enthalpy of atomisation of calcium.
- C: The second ionisation energy of calcium.
- D: The lattice energy of calcium chloride.

6 Lattice Energy & Born-Haber Variant 6 — 9701/4/23/Q6

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Increasing the interionic separation between cation and anion.
- D: Decreasing the ionic radius of the M²⁺ cation.

7 Lattice Energy & Born–Haber Variant 7 — 9701/4/23/Q7

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the ionic radius of the M²⁺ cation.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

8 Lattice Energy & Born–Haber Variant 8 — 9701/4/23/Q8

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Decreasing the ionic radius of the M²⁺ cation.
- D: Increasing the interionic separation between cation and anion.

9 Lattice Energy & Born–Haber Variant 9 — 9701/4/23/Q9

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the ionic radius of the M²⁺ cation.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

10 Lattice Energy & Born–Haber Variant 10 — 9701/4/23/Q10

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Decreasing the ionic radius of the M²⁺ cation.
- B: Increasing the ionic radius of the X⁻ anion.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

11 Lattice Energy & Born–Haber Variant 11 — 9701/4/23/Q11

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Increasing the interionic separation between cation and anion.
- D: Decreasing the ionic radius of the M²⁺ cation.

12 Lattice Energy & Born–Haber Variant 12 — 9701/4/23/Q12

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Decreasing the ionic radius of the M²⁺ cation.
- D: Increasing the interionic separation between cation and anion.

13 Lattice Energy & Born–Haber Variant 13 — 9701/4/23/Q13

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Decreasing the ionic radius of the M²⁺ cation.
- B: Increasing the ionic radius of the X⁻ anion.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

14 Lattice Energy & Born–Haber Variant 14 — 9701/4/23/Q14**HARD · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Decreasing the ionic radius of the M²⁺ cation.
- D: Increasing the interionic separation between cation and anion.

15 Lattice Energy & Born–Haber Variant 15 — 9701/4/23/Q15**EASY · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the ionic radius of the M²⁺ cation.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

16 Lattice Energy & Born–Haber Variant 16 — 9701/4/23/Q16**HARD · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Increasing the interionic separation between cation and anion.
- D: Decreasing the ionic radius of the M²⁺ cation.

17 Lattice Energy & Born–Haber Variant 17 — 9701/4/23/Q17**EASY · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Decreasing the ionic radius of the M²⁺ cation.
- D: Increasing the interionic separation between cation and anion.

18 Lattice Energy & Born–Haber Variant 18 — 9701/4/23/Q18**HARD · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Decreasing the ionic radius of the M²⁺ cation.
- B: Increasing the ionic radius of the X⁻ anion.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

19 Lattice Energy & Born–Haber Variant 19 — 9701/4/23/Q19**EASY · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Increasing the interionic separation between cation and anion.
- D: Decreasing the ionic radius of the M²⁺ cation.

20 Lattice Energy & Born–Haber Variant 20 — 9701/4/23/Q20**HARD · Syllabus 23.1**

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the ionic radius of the M²⁺ cation.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

21 Lattice Energy & Born–Haber Variant 21 — 9701/4/23/Q21

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the ionic radius of the M²⁺ cation.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

22 Lattice Energy & Born–Haber Variant 22 — 9701/4/23/Q22

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Increasing the interionic separation between cation and anion.
- D: Decreasing the ionic radius of the M²⁺ cation.

23 Lattice Energy & Born–Haber Variant 23 — 9701/4/23/Q23

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Decreasing the ionic radius of the M²⁺ cation.
- B: Increasing the ionic radius of the X⁻ anion.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

24 Lattice Energy & Born–Haber Variant 24 — 9701/4/23/Q24

HARD · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Increasing the ionic radius of the X⁻ anion.
- B: Decreasing the charge on the metal cation from +2 to +1.
- C: Decreasing the ionic radius of the M²⁺ cation.
- D: Increasing the interionic separation between cation and anion.

25 Lattice Energy & Born–Haber Variant 25 — 9701/4/23/Q25

EASY · Syllabus 23.1

Consider an ionic lattice MX₂. Which change causes the lattice energy to become MORE exothermic?

- A: Decreasing the ionic radius of the M²⁺ cation.
- B: Increasing the ionic radius of the X⁻ anion.
- C: Decreasing the charge on the metal cation from +2 to +1.
- D: Increasing the interionic separation between cation and anion.

SUBTOPIC 23.2 — ENTHALPIES OF SOLUTION & HYDRATION (Q26 – Q50)

26 Hess's Law for Enthalpy of Solution — 9701/11/M/J/22/Q2

EASY · Syllabus 23.2

Which equation correctly connects enthalpy of solution, lattice energy, and hydration enthalpies?

- A: $H^{\circ}\text{sol} = H^{\circ}\text{latt} - H^{\circ}\text{hyd}(\text{ions})$
- B: $H^{\circ}\text{sol} = H^{\circ}\text{latt} + H^{\circ}\text{hyd}(\text{ions})$
- C: $H^{\circ}\text{sol} = H^{\circ}\text{hyd}(\text{ions}) / H^{\circ}\text{latt}$
- D: $H^{\circ}\text{sol} = H^{\circ}\text{hyd}(\text{ions}) - H^{\circ}\text{latt}$

27 Group 2 Sulfate Solubility Trend — 9701/12/M/J/22/Q2

HARD · Syllabus 23.2

Why does the solubility of Group 2 sulfates decrease down the group from MgSO₄ to BaSO₄?

- A: Lattice energy decreases more rapidly than hydration enthalpy, making $H^{\circ}\text{sol}$ more exothermic.
- B: Hydration enthalpy decreases more rapidly than lattice energy, making $H^{\circ}\text{sol}$ increasingly endothermic.
- C: Lattice energy increases down the group due to increased molecular mass.
- D: Barium sulfate has higher covalent character than magnesium sulfate.

28 Group 2 Hydroxide Solubility Trend — 9701/13/M/J/22/Q2

HARD · Syllabus 23.2

Why does the solubility of Group 2 hydroxides increase down the group from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$?

- A: Hydration enthalpy decreases more rapidly than lattice energy.
- B: Hydroxide ion radius increases down the group.
- C: Lattice energy decreases more rapidly than hydration enthalpy, making H°_{sol} more exothermic.
- D: Barium hydroxide decomposes in water to form barium oxide.

29 Solution & Hydration Energetics Variant 29 — 9701/1/22/Q29

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^\circ_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

30 Solution & Hydration Energetics Variant 30 — 9701/1/22/Q30

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^\circ_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

31 Solution & Hydration Energetics Variant 31 — 9701/1/22/Q31

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^\circ_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The process is impossible and the solid cannot dissolve.
- D: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.

32 Solution & Hydration Energetics Variant 32 — 9701/1/22/Q32

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^\circ_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- D: The process is impossible and the solid cannot dissolve.

33 Solution & Hydration Energetics Variant 33 — 9701/1/22/Q33

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^\circ_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

34 Solution & Hydration Energetics Variant 34 — 9701/1/22/Q34

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- D: The process is impossible and the solid cannot dissolve.

35 Solution & Hydration Energetics Variant 35 — 9701/1/22/Q35

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

36 Solution & Hydration Energetics Variant 36 — 9701/1/22/Q36

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The process is impossible and the solid cannot dissolve.
- D: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.

37 Solution & Hydration Energetics Variant 37 — 9701/1/22/Q37

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- D: The process is impossible and the solid cannot dissolve.

38 Solution & Hydration Energetics Variant 38 — 9701/1/22/Q38

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

39 Solution & Hydration Energetics Variant 39 — 9701/1/22/Q39

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The process is impossible and the solid cannot dissolve.
- D: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.

40 Solution & Hydration Energetics Variant 40 — 9701/1/22/Q40

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

41 Solution & Hydration Energetics Variant 41 — 9701/1/22/Q41

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

42 Solution & Hydration Energetics Variant 42 — 9701/1/22/Q42

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The process is impossible and the solid cannot dissolve.
- D: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.

43 Solution & Hydration Energetics Variant 43 — 9701/1/22/Q43

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

44 Solution & Hydration Energetics Variant 44 — 9701/1/22/Q44

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- D: The process is impossible and the solid cannot dissolve.

45 Solution & Hydration Energetics Variant 45 — 9701/1/22/Q45

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

46 Solution & Hydration Energetics Variant 46 — 9701/1/22/Q46

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The process is impossible and the solid cannot dissolve.
- D: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.

47 Solution & Hydration Energetics Variant 47 — 9701/1/22/Q47

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

48 Solution & Hydration Energetics Variant 48 — 9701/1/22/Q48

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: Both the lattice energy and hydration enthalpies must be zero.
- C: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- D: The process is impossible and the solid cannot dissolve.

49 Solution & Hydration Energetics Variant 49 — 9701/1/22/Q49

EASY · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- B: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

50 Solution & Hydration Energetics Variant 50 — 9701/1/22/Q50

HARD · Syllabus 23.2

An ionic solid dissolves endothermically in water ($H^{\circ}_{\text{sol}} > 0$). What can be deduced about the magnitudes of lattice energy and hydration enthalpy?

- A: The magnitude of the lattice energy exceeds the sum of the hydration enthalpies of the ions.
- B: The sum of the hydration enthalpies exceeds the magnitude of the lattice energy.
- C: Both the lattice energy and hydration enthalpies must be zero.
- D: The process is impossible and the solid cannot dissolve.

SUBTOPIC 23.3 — ENTROPY CHANGE, S (Q51 – Q75)**51 Positive Entropy Change Identification — 9701/11/M/J/21/Q3**

EASY · Syllabus 23.3

Which reaction has a POSITIVE standard entropy change of the system, $S^{\circ}_{\text{sys}} > 0$?

- A: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$
- B: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
- C: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
- D: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

52 Entropy Change of Surroundings Formula — 9701/12/M/J/21/Q3

HARD · Syllabus 23.3

How is the entropy change of the surroundings, S_{surr} , related to the enthalpy change of the system, H_{sys} ?

- A: $S_{\text{surr}} = H_{\text{sys}} / T$
- B: $S_{\text{surr}} = -T \times H_{\text{sys}}$
- C: $S_{\text{surr}} = -H_{\text{sys}} / T$
- D: $S_{\text{surr}} = H_{\text{sys}} - TS_{\text{sys}}$

53 Units of Entropy — 9701/13/M/J/21/Q3

EASY · Syllabus 23.3

What are the correct SI units for standard molar entropy, S° ?

- A: $\text{J K}^{-1} \text{mol}^{-1}$
- B: kJ mol^{-1}
- C: J mol^{-1}
- D: $\text{kJ K}^{-1} \text{mol}^{-1}$

54 Entropy Analysis Variant 54 — 9701/1/21/Q54

HARD · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($S_{\text{sys}} < 0$) become spontaneous?

- A: The reaction is endothermic ($H > 0$) and the temperature is very high.
- B: The reaction can never be spontaneous under any conditions.
- C: The reaction is sufficiently exothermic ($H < 0$) and the temperature is sufficiently low.
- D: The temperature is above the boiling point of the solvent.

55 Entropy Analysis Variant 55 — 9701/1/21/Q55

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($S_{\text{sys}} < 0$) become spontaneous?

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56 Entropy Analysis Variant 56 — 9701/1/21/Q56

HARD · Syllabus 23.3

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57 Entropy Analysis Variant 57 — 9701/1/21/Q57

EASY · Syllabus 23.3

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58 Entropy Analysis Variant 58 — 9701/1/21/Q58

HARD · Syllabus 23.3

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59 Entropy Analysis Variant 59 — 9701/1/21/Q59

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

- A: The reaction is endothermic ($\Delta H > 0$) and the temperature is very high.
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- C: The temperature is above the boiling point of the solvent.
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60 Entropy Analysis Variant 60 — 9701/1/21/Q60

HARD · Syllabus 23.3

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61 Entropy Analysis Variant 61 — 9701/1/21/Q61

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62 Entropy Analysis Variant 62 — 9701/1/21/Q62

HARD · Syllabus 23.3

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63 Entropy Analysis Variant 63 — 9701/1/21/Q63

EASY · Syllabus 23.3

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64 Entropy Analysis Variant 64 — 9701/1/21/Q64

HARD · Syllabus 23.3

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65 Entropy Analysis Variant 65 — 9701/1/21/Q65

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

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66 Entropy Analysis Variant 66 — 9701/1/21/Q66

HARD · Syllabus 23.3

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- C: The temperature is above the boiling point of the solvent.
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67 Entropy Analysis Variant 67 — 9701/1/21/Q67

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

- A: The reaction is endothermic ($\Delta H > 0$) and the temperature is very high.
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68 Entropy Analysis Variant 68 — 9701/1/21/Q68

HARD · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

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69 Entropy Analysis Variant 69 — 9701/1/21/Q69

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

- A: The reaction is endothermic ($\Delta H > 0$) and the temperature is very high.
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70 Entropy Analysis Variant 70 — 9701/1/21/Q70

HARD · Syllabus 23.3

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71 Entropy Analysis Variant 71 — 9701/1/21/Q71

EASY · Syllabus 23.3

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- A: The reaction is endothermic ($\Delta H > 0$) and the temperature is very high.
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- C: The temperature is above the boiling point of the solvent.
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72 Entropy Analysis Variant 72 — 9701/1/21/Q72

HARD · Syllabus 23.3

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73 Entropy Analysis Variant 73 — 9701/1/21/Q73

EASY · Syllabus 23.3

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- D: The temperature is above the boiling point of the solvent.

74 Entropy Analysis Variant 74 — 9701/1/21/Q74

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- D: The temperature is above the boiling point of the solvent.

75 Entropy Analysis Variant 75 — 9701/1/21/Q75

EASY · Syllabus 23.3

Under which condition does a reaction with a negative entropy change ($\Delta S_{\text{sys}} < 0$) become spontaneous?

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- C: The reaction can never be spontaneous under any conditions.
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SUBTOPIC 23.4 — GIBBS FREE ENERGY CHANGE, ΔG (Q76 – Q100)

76 Condition for Thermodynamic Feasibility — 9701/11/O/N/21/Q4

EASY · Syllabus 23.4

What is the essential thermodynamic criterion for a reaction to be feasible at a given temperature?

- A: $\Delta H < 0$
- B: $\Delta S_{\text{sys}} > 0$
- C: $\Delta H > T\Delta S$
- D: $\Delta G \leq 0$

77 Calculation of Feasibility Temperature — 9701/12/O/N/21/Q4

HARD · Syllabus 23.4

A reaction has $\Delta H^\circ = +120 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = +150 \text{ J K}^{-1} \text{ mol}^{-1}$. Above what temperature does the reaction become feasible?

- A: 550 K
- B: 1250 K
- C: 800 K
- D: 0.80 K

78 Reaction Feasible at All Temperatures — 9701/13/O/N/21/Q4

EASY · Syllabus 23.4

Which combination of enthalpy and entropy changes guarantees that a reaction is thermodynamically feasible at ALL temperatures?

- A: $\Delta H < 0$ and $\Delta S > 0$
- B: $\Delta H > 0$ and $\Delta S < 0$
- C: $\Delta H < 0$ and $\Delta S < 0$
- D: $\Delta H > 0$ and $\Delta S > 0$

79 Linear Plot of ΔG against T — 9701/11/M/J/20/Q4

HARD · Syllabus 23.4

For the reaction $A(s) \rightarrow B(s) + C(g)$, a plot of ΔG against T is a straight line. What does the gradient of this line represent?

- A: $+\Delta S$
- B: ΔH
- C: $-\Delta H$
- D: $-\Delta S$

80 Gibbs Free Energy Feasibility Variant 80 — 9701/1/20/Q80

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

81 Gibbs Free Energy Feasibility Variant 81 — 9701/1/20/Q81

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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82 Gibbs Free Energy Feasibility Variant 82 — 9701/1/20/Q82

HARD · Syllabus 23.4

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83 Gibbs Free Energy Feasibility Variant 83 — 9701/1/20/Q83

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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84 Gibbs Free Energy Feasibility Variant 84 — 9701/1/20/Q84

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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- C: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- D: The reactants are completely insoluble in water.

85 Gibbs Free Energy Feasibility Variant 85 — 9701/1/20/Q85

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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- C: The total entropy change of the universe is negative.
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86 Gibbs Free Energy Feasibility Variant 86 — 9701/1/20/Q86

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87 Gibbs Free Energy Feasibility Variant 87 — 9701/1/20/Q87

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88 Gibbs Free Energy Feasibility Variant 88 — 9701/1/20/Q88

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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89 Gibbs Free Energy Feasibility Variant 89 — 9701/1/20/Q89

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
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- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

90 Gibbs Free Energy Feasibility Variant 90 — 9701/1/20/Q90

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- B: The reaction is endothermic and absorbs heat from the surroundings.
- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

91 Gibbs Free Energy Feasibility Variant 91 — 9701/1/20/Q91

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
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- C: The reactants are completely insoluble in water.
- D: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.

92 Gibbs Free Energy Feasibility Variant 92 — 9701/1/20/Q92

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

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- C: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- D: The reactants are completely insoluble in water.

93 Gibbs Free Energy Feasibility Variant 93 — 9701/1/20/Q93

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- B: The reaction is endothermic and absorbs heat from the surroundings.
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- D: The reactants are completely insoluble in water.

94 Gibbs Free Energy Feasibility Variant 94 — 9701/1/20/Q94

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The total entropy change of the universe is negative.
- C: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- D: The reactants are completely insoluble in water.

95 Gibbs Free Energy Feasibility Variant 95 — 9701/1/20/Q95

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

96 Gibbs Free Energy Feasibility Variant 96 — 9701/1/20/Q96

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The total entropy change of the universe is negative.
- C: The reactants are completely insoluble in water.
- D: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.

97 Gibbs Free Energy Feasibility Variant 97 — 9701/1/20/Q97

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The total entropy change of the universe is negative.
- C: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- D: The reactants are completely insoluble in water.

98 Gibbs Free Energy Feasibility Variant 98 — 9701/1/20/Q98

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- B: The reaction is endothermic and absorbs heat from the surroundings.
- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

99 Gibbs Free Energy Feasibility Variant 99 — 9701/1/20/Q99

EASY · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The total entropy change of the universe is negative.
- C: The reactants are completely insoluble in water.
- D: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.

100 Gibbs Free Energy Feasibility Variant 100 — 9701/1/20/Q100

HARD · Syllabus 23.4

A reaction has $G^\circ < 0$ at 298 K, but no reaction is observed when the reactants are mixed. What is the reason for this observation?

- A: The reaction is endothermic and absorbs heat from the surroundings.
- B: The activation energy for the reaction is too high, making the reaction kinetically unfeasible.
- C: The total entropy change of the universe is negative.
- D: The reactants are completely insoluble in water.

SUBTOPIC 23.1 — LATTICE ENERGY & BORN-HABER CYCLES (Q1 – Q25)

101 Core Repeat: Lattice Energy vs Ionic Charge & Radius — 9701/11/M/J/23/Q1

HARD · Syllabus 23.1

Which pair of ions will form the ionic lattice with the most exothermic lattice energy?

- A: Large cation with high charge and small anion with low charge
- B: Small cation with low charge and large anion with high charge
- C: Small cation with high charge and small anion with high charge
- D: Large cation with low charge and large anion with low charge

102 Core Repeat: Why Second Electron Affinity is Endothermic — 9701/12/M/J/23/Q2

HARD · Syllabus 23.1

Why is the process $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$ accompanied by an absorption of energy?

- A: The incoming electron is repelled by the negative charge of the O^- ion.
- B: An electron is promoted to a higher energy d-orbital.
- C: The effective nuclear charge increases dramatically.
- D: The atomisation enthalpy of oxygen is very high.

SUBTOPIC 23.2 — ENTHALPIES OF SOLUTION & HYDRATION (Q26 – Q50)

103 Core Repeat: Group 2 Sulfate Solubility Mechanism — 9701/13/M/J/23/Q3

HARD · Syllabus 23.2

Descending Group 2 from MgSO_4 to BaSO_4 , why does solubility decrease?

- A: Lattice energy decreases more rapidly than cation hydration enthalpy.
- B: The sulfate ion polarises the barium ion.
- C: The entropy of solution becomes significantly more positive.
- D: Cation hydration enthalpy decreases more rapidly than lattice energy.

104 Core Repeat: Group 2 Hydroxide Solubility Mechanism — 9701/11/O/N/23/Q3

HARD · Syllabus 23.2

Descending Group 2 from $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$, why does solubility increase?

- A: Hydration enthalpy decreases more rapidly than lattice energy.
- B: Lattice energy decreases more rapidly than cation hydration enthalpy.
- C: Barium hydroxide has covalent bonding character.
- D: The hydroxide ion is larger than the sulfate ion.

SUBTOPIC 23.3 — ENTROPY CHANGE, S (Q51 – Q75)

105 Core Repeat: Identifying $S > 0$ Processes — 9701/12/O/N/23/Q4

EASY · Syllabus 23.3

Which process is accompanied by an increase in entropy of the system?

- A: Sublimation of solid dry ice: $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$
- B: Condensation of steam: $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$
- C: Synthesis of sulfur trioxide: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$
- D: Freezing of molten copper: $\text{Cu}(\text{l}) \rightarrow \text{Cu}(\text{s})$

106 Core Repeat: Relationship Between ΔG and Spontaneity — 9701/13/O/N/23/Q4

EASY · Syllabus 23.3

According to the Second Law of Thermodynamics, a reaction is spontaneous if and only if:

- A: $\Delta G_{\text{sys}} > 0$
- B: $\Delta G_{\text{surr}} > 0$
- C: $\Delta G_{\text{tot}} = \Delta G_{\text{sys}} + \Delta G_{\text{surr}} > 0$
- D: $\Delta G_{\text{sys}} = \Delta G_{\text{surr}}$

SUBTOPIC 23.4 — GIBBS FREE ENERGY CHANGE, ΔG (Q76 – Q100)**107** Core Repeat: ΔG Feasibility Threshold Calculation — 9701/11/M/J/22/Q4

HARD · Syllabus 23.4

For a reaction with $H = +90 \text{ kJ mol}^{-1}$ and $S = +200 \text{ J K}^{-1} \text{ mol}^{-1}$, at what temperature does the reaction become feasible?

- A: 180 K
- B: 450 K
- C: 2.2 K
- D: 900 K

108 Core Repeat: Endothermic Feasible at High T — 9701/12/M/J/22/Q4

HARD · Syllabus 23.4

Under what thermodynamic conditions is a chemical reaction feasible only at HIGH temperatures?

- A: $H < 0$ and $S > 0$
- B: $H < 0$ and $S < 0$
- C: $H > 0$ and $S < 0$
- D: $H > 0$ and $S > 0$

109 Core Repeat: Exothermic Feasible at Low T — 9701/13/M/J/22/Q4

HARD · Syllabus 23.4

Under what thermodynamic conditions is a chemical reaction feasible only at LOW temperatures?

- A: $H < 0$ and $S < 0$
- B: $H > 0$ and $S > 0$
- C: $H < 0$ and $S > 0$
- D: $H > 0$ and $S < 0$

110 Core Repeat: Thermodynamic vs Kinetic Feasibility — 9701/11/O/N/22/Q4

HARD · Syllabus 23.4

Diamond is thermodynamically unstable relative to graphite at 298 K ($\Delta G^\circ = -2.9 \text{ kJ mol}^{-1}$ for diamond $\rightarrow$ graphite). Why does diamond not spontaneously convert to graphite?

- A: The entropy change for the conversion is negative.
- B: The reaction is strongly endothermic under standard conditions.
- C: The activation energy for breaking strong covalent C-C bonds in the giant diamond lattice is extraordinarily high.
- D: Graphite has a higher density than diamond.

Quick-Check Answer Key Grid — Topic 23 — Chemical Energetics (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 23 — Chemical Energetics (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

- 1 **B** Option B is correct. Lattice energy is defined specifically as the enthalpy change when 1 mole of an ionic crystalline solid is formed from its constituent separated gaseous ions under standard conditions (298 K, 100 kPa). It is always exothermic (negative).
- 2 **D** Option D is correct. Lattice energy is proportional to $(q_1 \times q_2) / (r_+ + r_-)$. MgO has ions with charges +2 and -2, and both Mg²⁺ (72 pm) and O²⁻ (140 pm) have small ionic radii. Compared to BaO (Ba²⁺ is much larger) and MgCl₂ (Cl⁻ has only -1 charge), MgO has the highest charge density and greatest electrostatic attraction, giving $H^\circ_{\text{latt}} = -3791 \text{ kJ mol}^{-1}$.
- 3 **A** Option A is correct. Adding an electron to an already negatively charged O⁻(g) ion experiences strong electrostatic repulsion. Work must be done against this repulsive force to bring the electron into the 2p subshell, making the second electron affinity strongly endothermic.
- 4 **C** Option C is correct. Theoretical lattice energies are calculated using the purely ionic electrostatic model. Ag⁺ has high polarising power (due to poor shielding by 4d¹⁰ electrons) and I⁻ has a large, easily distorted electron cloud. Polarisation results in electron sharing (covalent character), releasing extra energy and making the experimental lattice energy noticeably more exothermic.
- 5 **A** Option A is correct. CaCl₂ contains 2 moles of chloride ions per mole of formula units. Therefore, 2 moles of gaseous chlorine atoms must each gain an electron, requiring $2 \times \text{EA}(\text{Cl})$.
- 6 **D** Option D is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 7 **B** Option B is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 8 **C** Option C is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 9 **B** Option B is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 10 **A** Option A is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 11 **D** Option D is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 12 **C** Option C is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 13 **A** Option A is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 14 **C** Option C is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 15 **B** Option B is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 16 **D** Option D is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 17 **C** Option C is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 18 **A** Option A is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 19 **D** Option D is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 20 **B** Option B is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 21 **B** Option B is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 22 **D** Option D is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 23 **A** Option A is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 24 **C** Option C is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.
- 25 **A** Option A is correct. Decreasing the cation radius brings the oppositely charged ions closer together, increasing electrostatic attraction and making the lattice energy more exothermic.

26	D	Option D is correct. Applying Hess's Law across the dissolution cycle: dissolving solid into gaseous ions requires breaking the lattice ($-H^{\circ}\text{latt}$), and then hydrating the gaseous ions releases hydration enthalpy ($H^{\circ}\text{hyd}$). Thus $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$. [1]
27	B	Option B is correct. Down Group 2, cation radius increases. Since sulfate is a very large anion, lattice energy changes very little down the group. However, cation hydration enthalpy drops significantly. Because hydration energy drops much faster than lattice energy, $H^{\circ}\text{sol}$ becomes more endothermic (or less exothermic), causing solubility to decrease down the group. [1]
28	C	Option C is correct. OH^- is a small anion. As cation radius increases down Group 2, the internuclear separation increases significantly in percentage terms, causing lattice energy to decrease steeply. This steep decrease in lattice energy outweighs the drop in hydration enthalpy, making $H^{\circ}\text{sol}$ increasingly exothermic (or less endothermic) and thus increasing solubility down the group. [1]
29	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
30	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
31	D	Option D is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
32	C	Option C is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
33	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
34	C	Option C is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
35	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
36	D	Option D is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
37	C	Option C is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
38	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
39	D	Option D is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
40	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
41	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
42	D	Option D is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
43	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
44	C	Option C is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
45	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
46	D	Option D is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
47	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
48	C	Option C is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
49	B	Option B is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
50	A	Option A is correct. Since $H^{\circ}\text{sol} = H^{\circ}\text{hyd} - H^{\circ}\text{latt}$, if $H^{\circ}\text{sol}$ is positive (endothermic), then $ H^{\circ}\text{latt} > H^{\circ}\text{hyd} $, meaning more energy is required to break the crystal lattice than is released upon hydrating the ions. [1]
51	D	Option D is correct. A solid decomposes to produce a mole of gas (CO_2). Generating gas particles from a solid causes a massive increase in volume and number of accessible microstates, resulting in a large positive $S^{\circ}\text{sys}$. The other three options all involve a net decrease in gas moles or precipitation of ions into a solid lattice ($S < 0$). [1]

52	C	Option C is correct. When an exothermic reaction occurs ($H_{\text{sys}} < 0$), heat is transferred to the surroundings, increasing the thermal disorder and entropy of the surroundings: $S_{\text{surr}} = -H_{\text{sys}} / T$.	[1]
53	A	Option A is correct. Standard molar entropy is measured in Joules per Kelvin per mole ($\text{J K}^{-1} \text{mol}^{-1}$). Notice that enthalpy is typically given in kJ mol^{-1} , which requires a factor of 1000 conversion when calculating $G = H - TS$.	[1]
54	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
55	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
56	D	Option D is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
57	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
58	A	Option A is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
59	D	Option D is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
60	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
61	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
62	D	Option D is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
63	A	Option A is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
64	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
65	A	Option A is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
66	D	Option D is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
67	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
68	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
69	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
70	A	Option A is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
71	D	Option D is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
72	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
73	A	Option A is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
74	C	Option C is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
75	B	Option B is correct. For $G = H - TS$ to be negative when $S < 0$, H must be negative (exothermic) and T must be low enough so that $ H > TS $.	[1]
76	D	Option D is correct. A process is thermodynamically spontaneous/feasible if and only if the change in Gibbs free energy is negative or zero ($G \leq 0$), which is equivalent to the total entropy change being positive ($S_{\text{tot}} \geq 0$).	[1]
77	C	Option C is correct. At the feasibility boundary, $G = 0 = H - TS$. Therefore, $T = H / S = (120 \times 10^3 \text{ J mol}^{-1}) / (150 \text{ J K}^{-1} \text{ mol}^{-1}) = 800 \text{ K}$. Since both H and S are positive, the reaction is feasible at $T > 800 \text{ K}$.	[1]
78	A	Option A is correct. If H is negative and S is positive, then in $G = H - TS$, both H and $-TS$ are negative for any absolute temperature $T > 0 \text{ K}$. Thus G is always negative regardless of temperature.	[1]

79	D	Option D is correct. Comparing $G = -S(T) + H$ with the equation of a straight line $y = mx + c$: $y = G$, $x = T$, slope $m = -S$, and y-intercept $c = H$.	[1]
80	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
81	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
82	D	Option D is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
83	A	Option A is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
84	C	Option C is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
85	A	Option A is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
86	D	Option D is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
87	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
88	C	Option C is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
89	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
90	A	Option A is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
91	D	Option D is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
92	C	Option C is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
93	A	Option A is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
94	C	Option C is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
95	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
96	D	Option D is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
97	C	Option C is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
98	A	Option A is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
99	D	Option D is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
100	B	Option B is correct. A negative G indicates thermodynamic feasibility, but does not provide information about reaction rate. If the activation energy is high, the rate of reaction at room temperature is negligible, and the system remains kinetically stable.	[1]
101	C	Option C is correct. Electrostatic attraction in an ionic lattice is directly proportional to the product of ionic charges and inversely proportional to the sum of ionic radii. Maximum exothermic lattice energy requires maximum charge and minimum ionic radii.	[1]
102	A	Option A is correct. Work must be done to force a negatively charged electron onto an already negatively charged ion against electrostatic repulsion.	[1]
103	D	Option D is correct. The large size of SO_4^{2-} makes lattice energy nearly constant down the group, whereas hydration enthalpy drops sharply with increasing cation radius. Thus H°_{sol} becomes more endothermic, reducing solubility.	[1]
104	B	Option B is correct. Small OH^- anions allow lattice energy to drop steeply down Group 2 as cation radius increases, which outweighs the drop in hydration enthalpy, making H°_{sol} more exothermic.	[1]
105	A	Option A is correct. Conversion of a crystalline solid directly into freely moving gas particles represents a large increase in entropy.	[1]

106	C	Option C is correct. The Second Law states that the total entropy change of the universe (system + surroundings) must be positive for any spontaneous process.	[1]
107	B	Option B is correct. $T = H / S = (90 \times 10^3 \text{ J mol}^{-1}) / (200 \text{ J K}^{-1} \text{ mol}^{-1}) = 450 \text{ K}$.	[1]
108	D	Option D is correct. When $H > 0$ and $S > 0$, TS becomes larger than H only at sufficiently high temperatures, making $G = H - TS$ negative.	[1]
109	A	Option A is correct. When $H < 0$ and $S < 0$, the $-TS$ term is positive. Feasibility requires $ H > TS $, which holds only when T is low.	[1]
110	C	Option C is correct. Although thermodynamically feasible ($G^\circ < 0$), the reaction has an immense activation energy because it requires rupturing rigid tetrahedral covalent network bonds. Hence diamond is kinetically stable.	[1]